Two-Step Percentage Problems

In this lesson, you've learned that percents represent part of a whole. In many math problems and real-life situations, you must solve a percent equation, and then use the information to find another answer. The percent equation states that some percent of a whole equals the part.

Example 1

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Mario makes $14 an hour. After one year on the job, he gets a 7% raise. What is Mario's hourly wage after the raise?

Step 1

Set up the percent equation.

7% of $14 is *p*.

0.07 × 14 = *p*

Step 2

Solve for the unknown part.

0.07 × 14 = *p*

0.98 = *p*

Step 3

Decide what operation is needed for the second step

Add the increase to the old hourly pay to find the new hourly pay.

Step 4

Solve.

$0.98 + $14.00 = $14.98

Mario now earns **$14.98** per hour.

or in one step:

$p = (107\%)(14.00)$

$= (1.07)(14.00)$

$= \$14.98$

You can add a percent increase in one step by adding 100%. The raise was for 7%. That means the new hourly wage will be 107% of the current wage.

Example 2

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A jacket originally cost $120. The store put the jacket on sale for 20% off. When the jacket still didn't sell, the store reduced the price another 10%. What is the current sale price of the jacket?

Step 1

Find the amount of the first **discount**.

20% of $120 is *d*.

0.20 × $120 = $24

The first discount is $24.

Step 2

Subtract the first discount.

$120 − $24 = $96

$96 is the sale price after the first discount.

Step 3

Find the amount of the second discount.

10% of $96 is *d*.

0.10 × $96 = $9.60

The second discount is $9.60

Step 4

Subtract the second discount.

$96 − $9.60 = $86.40

The current price of the jacket is **$86.40.**

Example 3:

Zeke's rent increased by 5% last year, then by 3% this year. If his original rent was $1200.00 per month, what is his current rent?

(*i* = increase)

① $i_1 = (5\%)(1200.00)$

$= (0.05)(1200.00)$

$= \$60.00$ → so $r = 1200.00 + 60.00$

$= \$1260.00$

② $i_2 = (3\%)(1260.00)$

$= (0.03)(1260.00)$

$= \$37.80$

∴ The current rent is

$\$1260.00 + \$37.80$

$= \$1297.80$

After two discounts, the price of the jacket was reduced $24 + $9.60, for a total discount of $33.60. This is not the same as taking a 30% discount once. A 30% discount takes $36 off the price: $120 × .30 = $36. Always work the steps of a percent problem separately and in the order described in the problem.

Category
Lesson 7: Percentages
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